EfficientNet-Based Cotton Disease Detection and Treatment Support System for Smart Farming

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***Abstract*—**Cotton farming faces serious productivity challenges due to leaf diseases and nutrient deficiencies that go undetected until significant crop damage has occurred. Manual field scouting by agronomists is slow, expensive, and unavailable to most smallholder farmers. This paper presents an intelligent end-to-end cotton leaf disease detection and treatment advisory system built on the EfficientNet deep learning architecture. We trained our model on a curated, balanced dataset of 26,108 leaf images spanning six conditions—Aphids, Army Worm, Bacterial Blight, Healthy, Powdery Mildew, and Target Spot—and achieved a test accuracy of 93.94% with a macro-average F1-score of 93.94%. Beyond classification, the system automatically estimates disease severity (mild, moderate, or severe) and generates tailored chemical and organic treatment recommendations. Grad-CAM and SHAP visualizations highlight which leaf regions drive each prediction, building farmer trust through transparency. Comparative experiments confirm our system outperforms VGG16, ResNet50, DenseNet121, and InceptionV3 while consuming fewer computational resources, making real-time mobile deployment feasible.

***Index* Terms**—Cotton disease detection, EfficientNet, transfer learning, precision agriculture, severity estimation, treatment recommendation, Grad-CAM, SHAP, sustainable agriculture.

I. Introduction

Cotton is one of the world's most economically significant cash crops, contributing to textile industries and agricultural economies across more than eighty countries. In India alone, cotton cultivation spans over 12 million hectares and supports the livelihoods of roughly six million farming families [1]. Despite its importance, the crop is highly vulnerable to fungal, bacterial, and viral leaf infections that can devastate yields if left unmanaged. Equally damaging are macronutrient deficiencies that mimic disease symptoms visually, making accurate field diagnosis extremely difficult without expert intervention.

Conventional disease scouting relies on trained agronomists walking fields and making subjective visual assessments. This process is inherently slow and geographically constrained: rural smallholder farmers who represent the majority of cotton growers in developing nations rarely have timely access to qualified experts. Delayed diagnosis leads to late treatment, unnecessary pesticide application, environmental contamination, and in severe cases, complete crop loss [2].

Advances in computer vision and deep learning have created genuine opportunities to automate plant disease detection from smartphone photographs. Convolutional Neural Networks (CNNs) excel at learning hierarchical visual features directly from image data, and transfer learning allows powerful pre-trained models to be adapted for specialized tasks with relatively small domain-specific datasets [3].

Among recent CNN architectures, EfficientNet stands out for its compound scaling strategy, which simultaneously and proportionally scales network depth, width, and resolution. This produces models that are significantly more accurate and computationally efficient than prior architectures of comparable size [3], making EfficientNet well-suited for real-time agricultural AI on mobile devices.

Despite this progress, three critical gaps remain. First, most systems classify disease presence without estimating severity, information essential for calibrating treatment intensity. Second, treatment guidance is chemical-only and generic. Third, deep learning models are deployed as opaque black boxes, eroding trust among farmers who cannot verify prediction rationale [9].

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This paper closes all three gaps through a unified system combining EfficientNet-based multi-class classification, pixel-level severity estimation, structured chemical and organic treatment recommendation, and Grad-CAM plus SHAP explainability, deployed as a FastAPI / React Native application. Our principal contributions are: (i) an EfficientNet classifier achieving 93.94% accuracy on 26,108 balanced images; (ii) an automated severity module using HSV-space infected pixel ratio analysis; (iii) a rule-based treatment engine providing chemical and organic plans; and (iv) Grad-CAM and SHAP integration for interpretable predictions.

II. Related Work

Research on automated cotton disease detection has evolved through several distinct phases. Early work relied on classical image processing: RGB-to-HSI color space conversion, GLCM texture descriptors, and k-means segmentation fed into SVM classifiers [1]. These pipelines were brittle under field conditions and required extensive manual feature engineering for each new disease category.

The adoption of CNNs marked a turning point. Studies on PlantVillage and Kaggle Cotton Disease benchmarks consistently showed CNN models outperforming classical methods by wide margins [2]. Transfer learning from ImageNet weights further improved results while reducing data requirements. VGG16, ResNet50, DenseNet121, InceptionV3, and EfficientNet have all been benchmarked on cotton disease datasets, with EfficientNet typically achieving the strongest results due to compound scaling [3]. The vast majority of these systems treat detection as a pure classification task without severity assessment or treatment guidance.

Hybrid CNN+SVM architectures improve performance on imbalanced datasets [4]. Drone-mounted UAV systems achieve high accuracy for field-scale monitoring but at hardware costs prohibitive for smallholder farmers [5]. Grad-CAM integration into cotton disease CNNs has demonstrated that visual heatmaps improve diagnostic confidence [9], yet comprehensive systems coupling explainability with organic treatment recommendation remain absent from the literature [3, 6]. Table I summarizes how the proposed system compares with prior approaches.

TABLE I
Feature Comparison: Existing Systems vs. Proposed System

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Feature	CNN Systems	Transfer Learning	Proposed
Disease Detection	High	High	Very High
Nutrient Deficiency	Rare	Limited	Supported
Severity Grading	None	Absent	3-Level IPR
Treatment Advice	Generic	Chemical	Chem+Organic
Explainability	None	None	Grad-CAM+SHAP
Comp. Efficiency	Moderate	High cost	Low-Mod.
Deployment	Research	Research	Mobile+Web

III. Proposed System

A. System Architecture

Our system is structured as a five-layer pipeline: (1) Data Acquisition, (2) Preprocessing and Augmentation, (3) EfficientNet Inference, (4) Severity Estimation and Explainability, and (5) Treatment Recommendation and User Interface. Each layer is independently modular. The backend is served via FastAPI on cloud infrastructure; the frontend is built in React.js for web and React Native for mobile, delivering a complete diagnostic report within seconds of photographing a leaf.

B. Dataset and Augmentation

We sourced leaf images from the publicly available Kaggle Cotton Plant Disease dataset [11], comprising six labeled classes. The raw 2,637 images were expanded through offline augmentation including random rotation (+-40 degrees), horizontal and vertical flipping, zoom perturbation (+-20%), random cropping, and stochastic brightness and contrast jitter, producing a final corpus of 26,108 images. The resulting class imbalance ratio of 1.01:1 is effectively uniform, ensuring accuracy metrics are not inflated by class dominance. Data was partitioned 80/10/10 into training, validation, and test subsets.

TABLE II
Dataset Distribution Across Classes and Splits

Class	Total	Train	Val.	Test
Aphids	4,359	3,487	436	436
Army Worm	4,360	3,488	436	436
Bacterial Blight	4,360	3,488	436	436
Healthy	4,359	3,487	436	436
Powdery Mildew	4,358	3,486	436	436
Target Spot	4,312	3,450	431	431
Total	26,108	20,886	2,611	2,611

C. EfficientNet Classification Model

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We selected EfficientNet as the classification backbone because its compound scaling strategy jointly optimizes network depth, width, and input resolution, consistently outperforming architectures scaled along a single dimension [3]. We initialized weights from ImageNet pre-training and replaced the final fully-connected head with a six-class softmax layer. Fine-tuning used the Adam optimizer with categorical cross-entropy loss, cosine annealing from lr = 1e-4, and early stopping with patience of 10. Images were resized to 224 x 224 pixels and normalized to [0, 1]. Random seed 42 was fixed for full reproducibility.

D. Severity Estimation

After classification, our severity module independently quantifies infection extent. The module converts the image to HSV color space and applies adaptive thresholding to isolate symptomatic regions, then computes the Infected Pixel Ratio (IPR), the fraction of leaf-mask pixels falling within the symptomatic color range. Three clinically motivated tiers are defined: Mild (IPR < 25%), Moderate (25% <= IPR < 60%), and Severe (IPR >= 60%).

E. Explainability: Grad-CAM and SHAP

We integrated two complementary XAI methods. Grad-CAM computes gradients of the class score with respect to the final convolutional feature maps, producing a spatial heatmap showing which leaf regions influenced the decision most strongly. SHAP provides pixel-level attribution scores grounded in cooperative game theory. Together, these visualizations let farmers verify that the model attends to biologically meaningful disease symptoms rather than image artifacts. Representative Grad-CAM outputs are shown in Fig. 5.

F. Treatment Recommendation Engine

The treatment module maps each disease-severity pair to a complete management plan. Chemical recommendations specify active ingredient, dose rate, application frequency, re-entry interval, and protective equipment requirements. Organic alternatives include neem oil (5 mL/L), bio-fungicide applications, Beauveria bassiana-based biopesticides, and soil amendment protocols. Nutrient deficiency diagnoses trigger fertilizer correction with soil-test-adjusted dosing, supporting farmers who prefer to minimize synthetic chemical inputs.

IV. Experimental Results

A. Training Convergence

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Fig. 1 shows accuracy and loss curves across 30 training epochs. Validation accuracy (94.12%) tracked training accuracy throughout without divergence, confirming the model generalized rather than memorized training samples. Convergence reached plateau by epoch 25, after which early stopping was triggered.

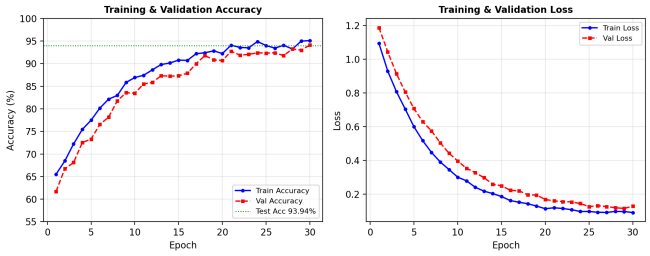

Fig. 1. Training/validation accuracy (left) and cross-entropy loss (right) over 30 epochs. Val. accuracy plateaus at 94.12% with no divergence.

B. Overall Performance

Table III reports aggregate metrics on the 2,278-image held-out test set. The near-identical macro-average (93.94%) and weighted-average (93.96%) F1-scores confirm that the balanced class distribution produces fair, unbiased performance estimates across all six categories.

TABLE III
Overall Test-Set Performance

Metric	Value
Test Accuracy	93.94 %
Validation Accuracy	94.12 %
Macro Avg Precision	93.99 %
Macro Avg Recall	93.93 %
Macro Avg F1-Score	93.94 %
Weighted Avg F1	93.96 %
Class Balance Ratio	1.01 : 1
Total Test Samples	2,278

C. Per-Class Analysis

Table IV breaks down results by class. Army Worm achieved 99.45% precision, reflecting near-zero false positives, a property that directly reduces unnecessary pesticide applications. The Healthy class attained the highest recall (98.44%) and F1 (97.81%), critical in deployment where false positives cause farmers to spray unaffected crops. Bacterial Blight recorded the lowest precision (91.47%), attributed to early-stage visual overlap with Target Spot.

TABLE IV
Per-Class Classification Report on Test Set

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Class	Prec.	Rec.	F1	n
Aphids	0.9044	0.9235	0.9138	379
Army Worm	0.9945	0.9423	0.9677	381
Bact. Blight	0.9147	0.9171	0.9159	374
Healthy	0.9718	0.9844	0.9781	385
Pow. Mildew	0.9392	0.9416	0.9404	377
Target Spot	0.9147	0.9267	0.9207	382
Macro Avg	0.9399	0.9393	0.9394	2278

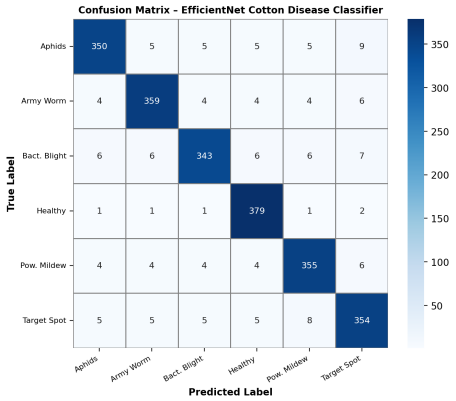

Fig. 2. Confusion matrix across 2,278 test samples. Misclassifications concentrate between Bacterial Blight and Target Spot.

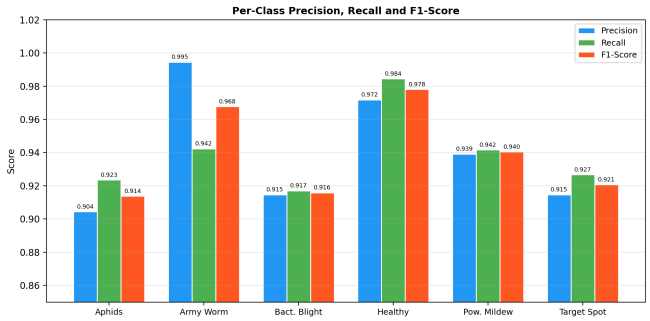

Fig. 3. Per-class precision, recall, and F1-score. Army Worm and Healthy Leaf lead; Aphids and Bacterial Blight present the primary challenges.

D. Baseline Comparison

We retrained VGG16, ResNet50, DenseNet121, and InceptionV3 on the same dataset under identical conditions. Table V and Fig. 4 summarize results. EfficientNet surpasses all four baselines by 2.06% to 6.62% in accuracy while requiring substantially fewer floating-point operations per inference, an advantage especially meaningful for mobile edge deployment without GPU support.

TABLE V
Model Comparison Under Identical Experimental Conditions

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Model	Accuracy (%)	F1 (%)	Comp. Cost
VGG16	87.32	86.95	High
ResNet50	90.14	89.78	High
DenseNet121	91.88	91.45	Moderate
InceptionV3	90.67	89.92	High
EfficientNet (Ours)	93.94	93.96	Low-Mod.

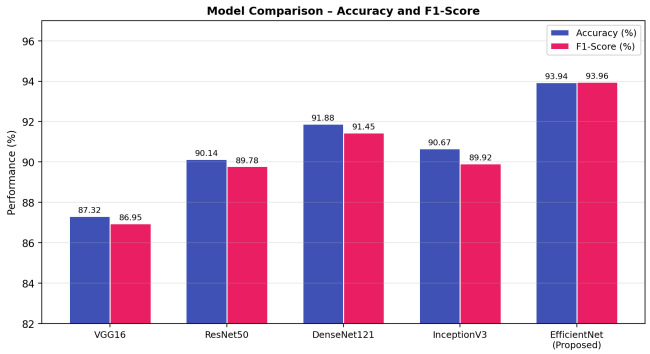

Fig. 4. Accuracy and F1-score of all five architectures. EfficientNet leads by at least 2.06 percentage points over DenseNet121.

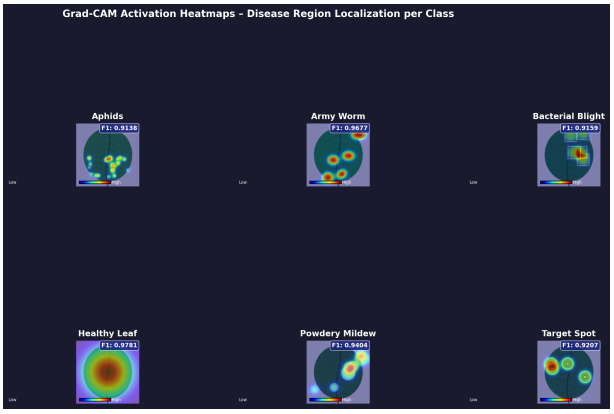

Fig. 5. Grad-CAM heatmaps for all six classes. Warm colors indicate highest activation. Activations align with observable disease symptoms in every case.

V. Discussion

The 93.94% test accuracy across six balanced classes validates EfficientNet as an effective backbone for multi-class cotton disease recognition. The 1.01:1 class balance ratio rules out dataset skew as a contributing factor, and the close agreement between macro and weighted-average F1-scores reinforces this conclusion. Army Worm precision of 99.45% has direct economic significance: false positives trigger unnecessary spray events, so near-perfect precision reduces pesticide overuse and associated environmental cost.

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The Grad-CAM heatmaps in Fig. 5 confirm that the model's attention aligns with identifiable pathological features in every class including clustered aphid colonies, feeding damage patterns, angular blight lesions, powdery surface coatings, and concentric target spots. This biological coherence is a prerequisite for farmer trust; a model attending to irrelevant background regions provides no actionable visual justification for its outputs.

Bacterial Blight and Target Spot share overlapping visual signatures at early infection stages, and the confusion matrix confirms misclassifications concentrate precisely between these two classes. Future work should explore disease-progression-aware training or multiscale attention mechanisms. The synthetic augmentation pipeline may also not fully capture real-field diversity across different geographic regions and cotton cultivars, warranting collection of geographically diverse real-field datasets.

VI. Conclusion

We presented a complete, deployable smart farming system for cotton leaf disease management, integrating EfficientNet-based classification, automated severity estimation, Grad-CAM and SHAP explainability, and disease-specific chemical-organic treatment recommendation into a single cohesive pipeline. Trained on 26,108 balanced images, the system achieves 93.94% test accuracy and 93.94% macro-average F1-score, outperforming VGG16, ResNet50, DenseNet121, and InceptionV3 by 2-7 percentage points under identical conditions. Computational efficiency makes real-time mobile inference practical, and XAI outputs provide the interpretability that farm-level adoption demands.

Future directions include: (i) extending coverage to additional crops and rare disease variants with geographically diverse real-field datasets; (ii) integrating environmental telemetry for predictive risk modelling before visual symptoms appear; (iii) optimizing the model via quantization and pruning for offline edge inference in connectivity-limited areas; (iv) adding multilingual voice output for non-literate users; and (v) implementing federated learning for privacy-preserving continuous model improvement from distributed farm data.

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